

CPSC 4970 – C++ Programming

Homework 4

Your solution to each problem should include the C++ source code and the screenshot of a test run. A solution without a test run will have points deducted.

1. Write a program that dynamically allocates an array large enough to hold a user-defined number of test scores. Once all the scores are entered, the array should be passed to a function that calculates the average score. At the end, the program should display the average score. Use pointer notation rather than array notation whenever possible.
2. Write a function that accepts an int array and the array's size as arguments. The function should create a copy of the array, except that the element values should be reversed in the copy. The function should return a pointer to the new array. Demonstrate the function in a complete program.
3. Write a function that accepts an int array and the array's size as arguments. The function should create a new array that is twice the size of the argument array. The function should copy the contents of the argument array to the new array and initialize the unused elements of the second array with 0. The function should return a pointer to the new array. Demonstrate the function in a complete program.
4. Write a function that generates and returns a dynamic two-dimensional array. The function should accept two integer arguments as the number of rows and the number of columns. The values in the 2D array should be randomly generated. Demonstrate the function in a complete program.